Priced In or Memorized?

Parts 7–8: Baselines, Model Comparisons and Financial Evaluation

Status report: what is built, what is missing, how to continue

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In one paragraph. The code for tasks 7 and 8 is written, commented, documented and tested (24 tests pass). It reads the outputs of parts 3–4, 5 and 6 directly, so new upstream data flows in by rerunning one command, with no code changes. On the real data, the surprise-only benchmark runs on the full sample (validation rank IC 0.31, $t = 22$). On synthetic data with a planted effect, the whole pipeline recovers the effect and rejects a pure-noise "LLM" and the placebo. What is missing is mostly *upstream input*: release texts beyond 2006 (part 5), the Loughran–McDonald dictionary file, model scores (part 6), and the holdout freeze. A handful of design decisions must also be written down before the freeze.

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1 What parts 7 and 8 are for

The project asks whether the text of an earnings release tells us anything about the stock's abnormal return that the earnings surprise does not, and whether language models do this better than simple text models, or only appear to because they memorised outcomes.

- **Task 7** builds the simple benchmarks a language model has to beat (surprise only, Loughran–McDonald tone, TF-IDF) and compares every model on the same sample, with the same metrics, windows and chronological splits.
- **Task 8** merges model scores with the event-level financial data and runs the tests: regressions controlling for the surprise and firm characteristics, a 2×2 table (surprise sign $\times$ text sign), robustness checks, and the 2022 rate shock.

The four claims the design is built to test, in increasing order of strength:

	Claim	Test
A	Text is related to returns	Rank IC, long-short spread
B	Text adds information *beyond the surprise* (the core claim)	Text coefficient with SUE deciles; combined vs surprise-only out of sample (Clark–West); 2×2 table
C	LLMs beat simple text models	Same comparison: LLM vs Loughran–McDonald vs TF-IDF
D	It is real, not an artefact	Robustness grid, day -1 placebo, one run on the frozen holdout

2 How it plugs into the rest of the project

Part	What it provides	How 7–8 reads it
3–4	Event table: release clock, point-in-time consensus, SUE, abnormal returns (FF4 and market-adjusted), splits	`3_4_surprise_returns/data/events_final.parquet`; its `common.py` and `returns.py` are imported, not copied
5	Cleaned release texts; frozen holdout ids	`5_data cleaning/data/clean/clean_*.parquet`; `holdout_ids.txt`
6	Model scores (FinBERT, ChronoGPT, Llama 3.1 8B)	`5_data cleaning/cache/model_outputs.sqlite`, read-only; each `(model, prompt, variant)` becomes a signal `llm__<model>__<prompt>__<variant>`
9	Memorisation tests	Uses the Llama-cutoff split and the rate-shock regimes produced here
10	Report	Every table is saved as `.csv`, `.md` and `.tex`

Column contracts are checked at load time (`io_inputs.py`); a missing column fails loudly rather than producing a wrong number. Full contracts are in `docs/INTERFACES.md`.

3 Design in brief

The rationale is in `docs/METHODOLOGY.md`; the essentials are:

1. **A strong surprise benchmark.** SUE enters as decile dummies because the return response is S-shaped. A linear benchmark would let text models win by capturing that curve.

2. **Every model graded the same way.** One panel of events. Each signal is converted to its within-quarter rank in $[-0.5, 0.5]$, so a 1–5 LLM grade, a dictionary tone and a TF-IDF prediction are comparable. Comparisons use *common support*: only events every model scored.
3. **Nothing sees the future.** Trained models are fitted on earlier years only (expanding window), with a 100-day embargo so a training label's 60-day return never overlaps the test period. Tuning uses the development years only. The holdout (2021–25) needs two keys to open. A day -1 placebo catches timing errors.
4. **Inference respects clustering.** Releases cluster in quarterly seasons that share market news, so statistics are computed within season and seasons are the independent units (Newey–West, 2 lags). Regressions use two-way clustering by firm and season, with Fama–MacBeth as a check. Validation has only 12 seasons: below $|t| = 3$ we say "suggestive".

Split	Years	Role	Status
dev	2006–2016	Fit and tune	In use
validation	2017–2019	Comparison set	In use
excluded	2020	COVID year, never used	–
holdout	2021–2025	Final test, one run	Locked until freeze

4 What is built: task 7

Requirement	Implementation	Status
Surprise-only baseline	`baseline_surprise.py`: SUE rank, linear and decile models, cross-fitted	**Done** (real data)
Loughran–McDonald baseline	`baseline_lm.py`: 7 categories, negation of positive words within 3 words, signals `lm_tone`, `lm_net`, `lm_negshare`, `lm_unc`; counts cached by text hash	**Done** (code); **Waiting** dictionary file
TF-IDF baseline	`baseline_tfidf.py`: uni/bigrams, numbers masked, firm filter (≥ 20 firms per term), ridge and logistic, tuned on dev only	**Done** (code); **Waiting** texts
Language models on same sample	`io_inputs.load_model_scores`: reads part 6's cache, `raw` and `mask` variants	**Done** (plumbing); **Waiting** scores
Surprise-only vs text-only vs combined	`compare.py`: out-of-sample R^2, gain over surprise-only, Clark–West with t_{S-1} p-values	**Done**
Consistent metrics, windows, splits	`metrics.py`, `crossfit.py`: same IC, long-short, hit rate, AUC, five windows, one split scheme	**Done**
Complex vs simple models	Combined-model rows for every signal in one table	**Done**; needs LLM scores

5 What is built: task 8

Requirement	Implementation	Status
Merge scores with financial data	`panel.py` + `pipeline.py`: one row per event, signals aligned by event	**Done**
Signal–return relationship	IC and decile long-short per signal and window	**Done**
Regressions with controls	`regressions.py`: ladder (1)–(5), controls (size, analysts, dispersion, prior return), FF12 sector effects, two-way clustering, Fama–MacBeth	**Done**
2×2 table	`two_by_two.py`: beat/miss × good/bad tone (within-season median), cell means, SEs, 95% CIs, n, contrasts incl. diff-in-diff	**Done**
Text beyond surprise	Spec (3) coefficient, Clark–West, 2×2 within-row effects	**Done**
Robustness	`robustness.py`: sectors, size, NYSE, periods, Llama cutoff, all windows, placebo, stricter samples, dispersion-scaled SUE	**Done**
2022 rate shock	`rate_shock.py`: FOMC-dated regimes, per-regime IC and coefficient, pooled interaction + Wald test	**Done**; **Locked** (holdout)
Save outputs automatically	`outputs.py`: `.csv/.md/.tex`, provenance log (sha256, git commit, upstream versions), refuses row-level tables	**Done**

Quality checks

- **24 tests** (`python code/tests/run_all.py`): hand-computed answers (ranks, Newey–West equal to part 4's, Clark–West, exact 2×2 cell means, LM negation, folds without look-ahead, holdout lock) and recovery tests on synthetic data (planted effect found; noise and placebo not; Clark–West size under the null over 15 seeds).
- **Notebooks**: `00_inputs_check`, `07_baselines`, `08_evaluation`, `08b_rate_shock`, each with markdown explaining the choices.

6 Results so far

6.1 Real data: the panel

Split	Events	With text	Seasons	First	Last
dev	35,508	105	44	2006-01-04	2016-12-22
validation	11,375	10	12	2017-01-04	2019-12-20

Counts match part 4's `sample_by_split.csv`. Only 115 events have text so far.

6.2 Real data: the surprise benchmark (validation, 2017–19)

Outcome	Rank IC	t_{NW}	Long-short	t_{NW}	AUC
Reaction, `car_0_1_ff4`	0.314	22.3	7.52%	13.3	0.655
`post_car_1_ff4`	0.026	2.5	0.22%	0.6	0.514
Drift, `post_car_60_ff4`	−0.011	−0.7	−1.35%	−1.9	0.497
`car_1_60_ff4`	−0.020	−1.3	−1.66%	−2.4	0.495
Placebo, `ar_m1`	0.036	4.7	0.26%	4.5	0.521

Reading: the surprise is strongly priced at release and shows no post-announcement drift in 2017–19. It also predicts the day −1 return, consistent with pre-announcement drift (leakage or anticipation). That is why the placebo is reported *controlling for SUE*: a text signal only fails the placebo if it predicts day −1 beyond what the surprise does. The out-of-sample R^2 of the surprise-only decile model on the reaction is 8.7%: this is the bar text models have to raise.

6.3 Synthetic data: the pipeline works

Synthetic events carry a planted text effect (a "good" LLM) and a pure-noise LLM. Combined model vs surprise-only, reaction window, validation split:

Text signal	Text-only IC	OOS R^2 gain	Clark–West t	p
Good LLM	0.314	+3.8 pp	17.1	$< 10^{-8}$
LM tone	0.295	+3.0 pp	10.2	$< 10^{-6}$
TF-IDF ridge	0.259	+3.0 pp	6.0	$< 10^{-4}$
Noise LLM	0.010	+0.0 pp	0.9	0.19

These numbers are not findings; they show the machinery finds an effect that exists and does not invent one that does not.

7 What is missing

7.1 Upstream inputs (blocking)

Item	Who	Effect on 7–8	Action here
Release texts 2007–2025	Part 5	LM and TF-IDF cover 115 events; TF-IDF needs ≥ 500 training texts per fold	Rerun `--step all`
Loughran–McDonald dictionary	Anyone (sraf.nd.edu)	LM baseline skipped	Put the CSV in `resources/lm/`
LLM scores	Part 6	No LLM rows in any table	Rerun `--step evaluate`
Corrected release clock	Part 1, then 3–4	Returns and surprise may shift	Rerun `--step all`
Holdout freeze	Part 5 (`splits.py --freeze`)	Test run and rate shock locked	Final run (§8)

7.2 Decisions to write down before the freeze

These must be fixed in `docs/METHODOLOGY.md` (ideally agreed with the instructor) before the holdout is opened; afterwards they cannot change.

1. **Headline outcome.** Proposed: `car_0_1_ff4` (reaction) as primary; the 60-day post-release window as secondary.
2. **Main sample.** `in_primary` (3+ analysts) vs the broader sample; with or without the `basis_suspect` and `timing_suspect` events.
3. **Main LLM signal.** Which model, prompt and variant (`raw` vs `mask`) is the headline in the 2×2 table and the regressions. The others become robustness rows.
4. **Size filter.** Whether to drop micro-caps (e.g. below the NYSE 20th percentile).
5. **Multiple testing.** How many signals $\times$ windows are "primary"; the rest are labelled exploratory.
6. **Interpretation table.** What each pattern of results would mean (priced in vs under-reaction vs memorisation), written before seeing the holdout.

7.3 Extensions not yet built (own work, optional)

- **Tradable portfolio.** Current long-short uses same-quarter ranks (fine for measuring information, not a strategy). A version with breakpoints from trailing releases only would give an implementable return.
- **Book-to-market control.** Needs Compustat `ceqq` from part 3–4; then one line in `regressions.CONTROLS`.
- **FactSet consensus robustness.** If part 3 provides it, add an alternative `sue_*` column to the robustness grid.
- **Figures for the report.** Cumulative abnormal return by text tercile within beat/miss (event-time plot), IC by season over time.
- **Initial git commit.** The repo is initialised and staged but not committed.

8 How to continue

All commands run from the folder `7_8_baselines_evaluation/`.

1. **Now:** download the Loughran–McDonald master dictionary into `resources/lm/`, then `python code/pipeline.py --step all`. Check `results/tables/`.
2. **When part 5 delivers more texts:** rerun `--step all`. Sanity-check the TF-IDF top terms (they should be words like "impairment", "raised guidance", not firm names), text coverage by year, and the placebo row.
3. **When part 6 fills the model cache:** `python code/pipeline.py --step evaluate`. LLM signals appear automatically in every table.
4. **Make the decisions above** and record them in `docs/METHODOLOGY.md` and `docs/CHANGELOG.md`.

5. **After the freeze, once:**
   `UNLOCK_HOLDOUT=1 python code/pipeline.py --step all --eval-split holdout`, then run `08b_rate_shock.ipynb`.
6. **Hand over to part 10:** the `.tex` tables in `results/tables/` can be `\input` directly into the report.

What each upstream change requires

Upstream change	Rerun
Part 5 cleaned more texts	`07_baselines`, then `08_evaluation` (LM counts are cached)
Part 6 added model scores	`08_evaluation` only
Part 1/3/4 corrected clock, surprise or returns	Everything (`--step all`)
Holdout frozen	The single final run

A File map

File	Role
`code/settings.py`	Paths and research-design constants (outcomes, splits, dates, seeds)
`code/io_inputs.py`	Readers for parts 3–4, 5, 6 with column contracts
`code/panel.py`	Analysis panel: seasons, ranks, controls, winsorising, holdout guard
`code/crossfit.py`	Chronological out-of-sample fitting with embargo
`code/baseline_surprise.py`	Benchmark 1: SUE
`code/baseline_lm.py`	Benchmark 2: Loughran–McDonald
`code/baseline_tfidf.py`	Benchmark 3: TF-IDF + ridge / logistic
`code/metrics.py`	IC, long-short, OOS R^2, Clark–West, hit rate, AUC
`code/compare.py`	Surprise-only vs text-only vs combined
`code/regressions.py`	Ladder, two-way clustering, Fama–MacBeth
`code/two_by_two.py`	2×2 table and contrasts
`code/robustness.py`	Robustness grid
`code/rate_shock.py`	2022 rate-shock regimes
`code/outputs.py`	Saving and provenance logging
`code/synthetic.py`	Synthetic data with planted effect
`code/pipeline.py`	Command-line entry point
`docs/METHODOLOGY.md`	Why everything is done this way
`docs/INTERFACES.md`	Column and score contracts with other parts

Data note. This report contains summary statistics only. Row-level WRDS derivatives stay in `data/` on the team Drive and are never committed or shared.